Prime Factorization

Flagship

Lesson 1-1-flagship

Name: ____________ Date: ____________ Class: ____________

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

7 has only two factors: 1×7. So 7 is prime.

Prime number

A number bigger than 1 that you can only divide by 1 and itself.

$12 = 1 \times 12$, 2×6, 3×4 — six factors, so 12 is composite

Composite number

A number bigger than 1 that you can divide by more than just 1 and itself.

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Prime factorization

Writing a number as prime numbers multiplied together.

$24 \rightarrow 4 \times 6 \rightarrow (2 \times 2) \times (2 \times 3) \rightarrow 2 \times 2 \times 2 \times 3$

Factor tree

A picture that splits a number into its prime numbers, step by step.

2^3 means $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply a number by itself.

Key Ideas & Notes

- The space station received 60 supply crates.
- Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods.
- Help the crew find the prime factorization of 60!
- Sort these numbers — which are prime and which are composite?

Think About It

- What number are we breaking down into factors?
- What's the difference between a factor and a prime factor?
- How many different ways can we start breaking 60 apart?

My Notes

Turn & Talk — Discussion Points

Level 1 support · Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The freighter docks with 48 cargo containers, and each storage pod only holds groups built from prime-numbered units. Is 48 prime or composite, and how can you tell before building a factor tree?

Level 1 support

Start here: 48 is composite because it has more than two factors.

48 is a ____ number because it has ____.

48 es un número ____ porque tiene ____.

I can tell before factoring because ____.

Puedo saberlo antes de factorizar porque ____.

Word bank: prime number · composite number · factor · divisible · factor tree

Listen for: *Students say 48 is composite because it has more than two factors (divisible by 2, 3, 4, 6, ...), so it can be broken into smaller groups.*

Level 2

Why can a prime-numbered shipment, like 47 containers, NOT be split into smaller equal prime groups? Justify with the definition of prime.

- 47 cannot be split because its only factors are ____.
- A prime number's factor tree would ____ because ____.

EXPLORE

Look at your factor tree for 48. How did you decide which numbers to break apart first, and how did you know when to stop?

Level 1 support

Start here: You keep splitting until every branch is a prime number.

I broke 48 into ___ first because ___.

Separé 48 en ___ primero porque ___.

I knew I was finished when every factor was ___.

Supe que terminé cuando cada factor era ___.

Word bank: **prime number** **composite number** **factor** **factor tree** **prime factorization**

Listen for: *Students explain they keep splitting composite branches and stop when every leaf is prime, ending with* $48 = 2 \times 2 \times 2 \times 2 \times 3$.

Level 2

Write the prime factorization of 48 using exponents and explain what the exponent means in $2^4 \times 3$.

- The prime factorization of 48 with exponents is ___.
- The exponent in 2^4 means ___ because ___.

CONNECT

The cafeteria has 84 apples to arrange into equal rows. How does the prime factorization of 84 help the manager find every possible equal-row display?

Level 1 support

Start here: Combining the prime factors of 84 builds all of its factors.

The prime factorization of 84 is ___.

La factorización prima de 84 es ___.

The manager finds all the row sizes by ___ the prime factors.

El gerente halla todos los tamaños de fila al ___ los factores primos.

Word bank: **prime factorization** | **factor** | **exponent** | **composite number** | **product**

Listen for: *Students give* $84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$ *and reason that multiplying combinations of those primes produces all factors of 84, which are the possible equal-row sizes.*

Level 2

Why does knowing the prime factorization of 84 make it faster to list all factors than checking every number from 1 to 84?

- Prime factorization is faster because ___.
- I can build factors by ___ instead of testing every number since ___.

REFLECT

Why does every composite number have exactly one prime factorization, no matter how you start the factor tree?

Level 1 support

Start here: Primes are unique building blocks, so the set of prime factors is always the same.

No matter how I start the tree, I always end with ___ because ___.

No importa cómo empiece el árbol, siempre termino con ___ porque ___.

Prime numbers are like building blocks because ___.

Los números primos son como bloques de construcción porque ___.

Word bank: **prime number** **composite number** **prime factorization** **factor tree** **product**

Listen for: *Students explain that breaking down different starting splits still reduces to the same prime factors, so the prime factorization is unique (the Fundamental Theorem of Arithmetic in plain words).*

Level 2

Start 48 two different ways (such as 6×8 and 4×12) and show both give $2^4 \times 3$. Explain why this had to happen.

- Starting with 6×8 gives ___, and starting with 4×12 gives ___.
- Both match because ___.

Guided Examples

Example 1

Which of the following is a prime number?

Solution: 17 has exactly two factors: 1 and 17. $15 = 3 \times 5$, $21 = 3 \times 7$, and $9 = 3 \times 3$, so they are all composite.

Answer: A. 17

Example 2

What is the prime factorization of 30?

Solution: $30 = 2 \times 15 = 2 \times 3 \times 5$. All three factors (2, 3, 5) are prime, so $2 \times 3 \times 5$ is the prime factorization.

Answer: A. $2 \times 3 \times 5$

Example 3

What is the prime factorization of 18?

Solution: $18 = 2 \times 9 = 2 \times 3 \times 3$. Both 2 and 3 are prime, so $2 \times 3 \times 3$ is the prime factorization.

Answer: A. $2 \times 3 \times 3$

Write About the Math

The Writing Revolution

I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

1. Kernel Sentence

subject + verb

Model: Prime factorization is writing a number as prime numbers multiplied together.
Factorización prima es escribir un número como números primos multiplicados.

Write a kernel sentence about prime factorization. Use a subject and a verb.
Escribe una oración base sobre factorización prima. Usa un sujeto y un verbo.

2. Sentence Expansion

because · but · so

Kernel: Prime factorization matters in math
Factorización prima importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Prime factorization matters in math because ____.
Factorización prima importa en matemáticas porque ____.

but
pero

Prime factorization matters in math, but ____.
Factorización prima importa en matemáticas, pero ____.

so
entonces

Prime factorization matters in math, so ____.
Factorización prima importa en matemáticas, entonces ____.

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about prime factorization.
Di un hecho verdadero sobre prime factorization.

Prime factorization ___.

Question
Pregunta

Ask a question about prime factorization.
Haz una pregunta sobre prime factorization.

How does ___ ?
¿Cómo ___ ?

Exclamation
Exclamación

Show excitement about prime factorization.
Muestra entusiasmo sobre prime factorization.

Wow, ___ !
¡Guau, ___ !

Command
Mandato

Tell a partner what to do with prime factorization.
Dile a un compañero qué hacer con prime factorization.

First, ___ .
Primero, ___ .

4. Explain Your Reasoning

use a sentence starter

48 is a ___ number because it has ___.
48 es un número ___ porque tiene ___.

I can tell before factoring because ___.
Puedo saberlo antes de factorizar porque ___.

I broke 48 into ___ first because ___.
Separé 48 en ___ primero porque ___.

Try It

Solve on your own. Check the answer key when you are done.

1. Which of these numbers is composite?

A. 27

B. 23

C. 29

D. 31

Show your work:

2. Two students found different factor trees for 60. Student A started with 2 × 30. Student B started with 6 × 10. Which statement is true?

A. Both get the same prime factorization: 2 × 2 × 3 × 5

B. Only Student A gets the correct prime factorization

C. Only Student B gets the correct prime factorization

D. They will get different prime factorizations

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Choose any two-digit composite number. Show TWO different factor trees that both lead to the same prime factorization. Explain why every composite number has only one prime factorization.

Sentence starter: I chose the number ____. My first factor tree starts with ____ × ____, and my second starts with ____ × ____. Both give the same prime factorization: ____. This happens because ____.

Show your work:

Reflect — Exit Ticket

What is the prime factorization of 40?

A. 2 × 2 × 2 × 5

B. 4 × 10

C. 5 × 8

D. 2 × 20

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 27 *— 27 = 3 × 9 = 3 × 3 × 3, so it has more than two factors. 23, 29, and 31 are all prime.*
2. **Try It 2:** A. Both get the same prime factorization: 2 × 2 × 3 × 5 *— The Fundamental Theorem of Arithmetic says every composite number has exactly one prime factorization. No matter how you start the factor tree, you always end with 2 × 2 × 3 × 5.*
3. **Exit Ticket:** A. 2 × 2 × 2 × 5 *— 40 = 2 × 20 = 2 × 2 × 10 = 2 × 2 × 2 × 5. All factors (2, 2, 2, 5) are prime.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Prime factorization is writing a number as prime numbers multiplied together.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).